

An Amphibious Magnetotelluric Experiment at the South Chilean Continental Margin

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Abstract

A combined on- and offshore magnetotelluric measurement was carried out in the South Chilean continental margin to resolve the crustal and upper mantle distribution of electrical conductivity and in particular to image fluids between the downgoing Nazca and overriding South American plates. This contribution presents the MT data and some specific problems encountered in this sea-land experiment.

Introduction

Previous long-period magnetotelluric studies in the Southern Chilean Andes revealed a modest high conductivity zone (HCZ) in the lower crust beneath the active volcanic arc and an additional HCZ associated with the Lanalhue fault in the forearc, running obliquely to the main morpho-structural units (Brasse and Soyer 2001, Soyer 2002, Melnick et al. 2006). Another finding is based on the anomalous behavior of geomagnetic induction vectors, which point consistently NE over the whole study area instead of the expected E direction due to the presence of the highly conducting Pacific Ocean. This spectacular effect is (and can only be) explained by anisotropy of the crust.

To corroborate these unexpected results and especially to image fluids in the interface between the downgoing Nazca and the overriding South American plate a combined on-offshore electromagnetic experiment was carried out as part of the multi-disciplinary TIPTEQ project. 33 long-period magnetotelluric stations measured the variation of magnetic and electric fields from December 2004 to the end of February 2005 between latitudes 37.5° - 41°S. 23 stations on the land side and 7 stations on the sea side were established along a profile with a separation of approx. 10 km and 30-35 km respectively, extending over 400 km and running perpendicularly to the trench and the volcanic arc from the incoming plate to the Argentinian border.

The profile is a part of a large-scale magnetotelluric network including 3 profiles in this area, several connecting sites and small networks of stations around the active volcanoes Villarrica and Llaima to study the conductivity distribution and to get indications of possibly existing deep magma chambers, respectively.

For monitoring of temporary variability in the geomagnetic transfer function, 3 stations from GFZ Potsdam have been carried out close to the profile and on Isla Mocha for a time period of 3 years.

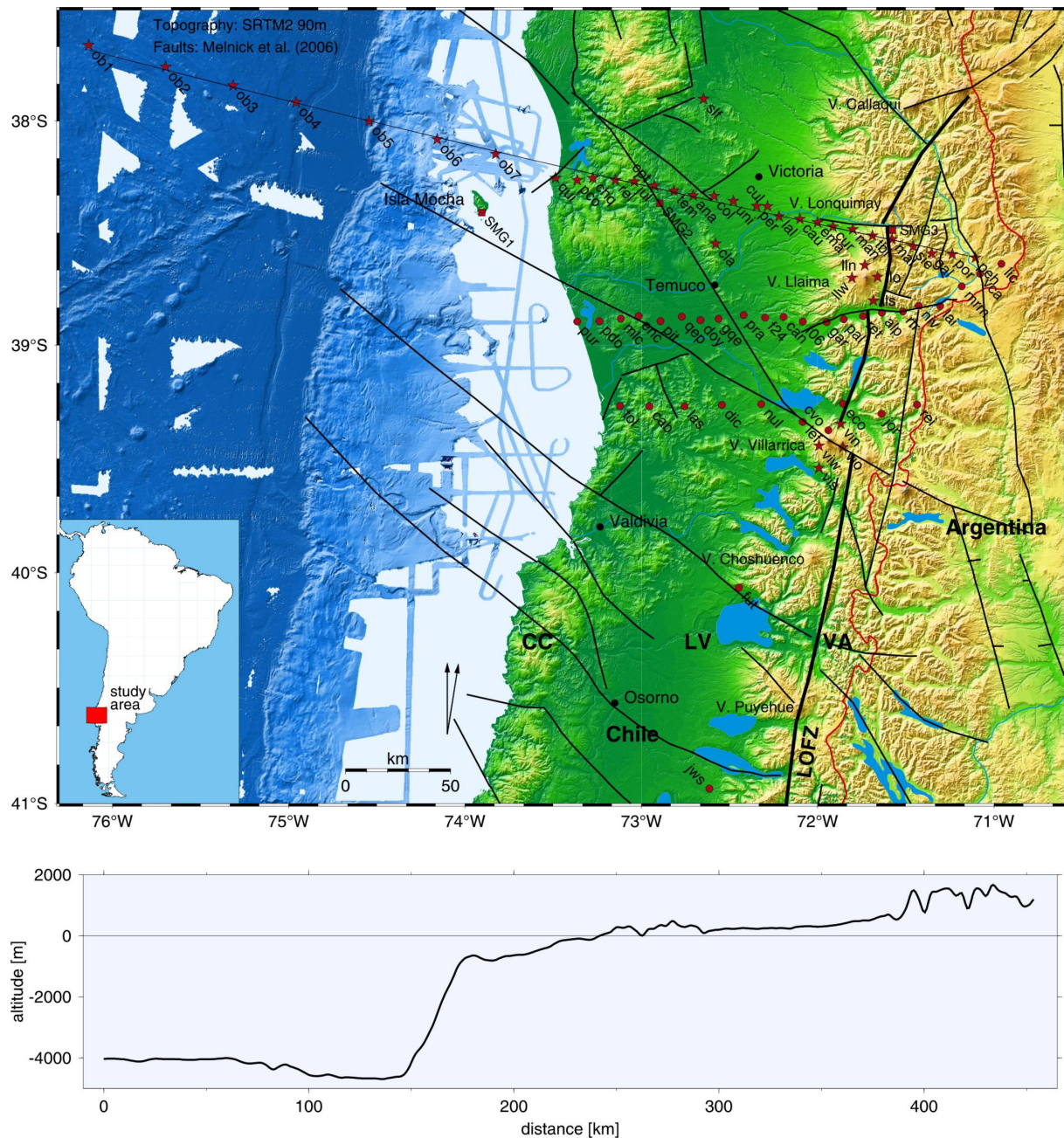


Fig. 1: Map with MT sites at the South Chilean margin. Circles: Sites from campaign in 2000; Stars: TIPTEQ campaign; Squares: Monitoring stations from GFZ Potsdam; Black lines: Faults modified after Melnick et al. 2006; Bathymetry: GEOMAR. CC: Coastal Cordillera, LV: Longitudinal Valley, VA: Volcanic Arc, LOFZ: Liquiñe-Ofqui Fault Zone. Below: Bathymetry/topography along magnetotelluric profile.

Ocean-bottom magnetotellurics

For imaging fluids in the interface between the downgoing Nazca and the overriding South American plate the last land profile was extended by 7 sea-bottom stations across the trench towards the Pacific plate, which were deployed during RV Sonne leg SO-181.

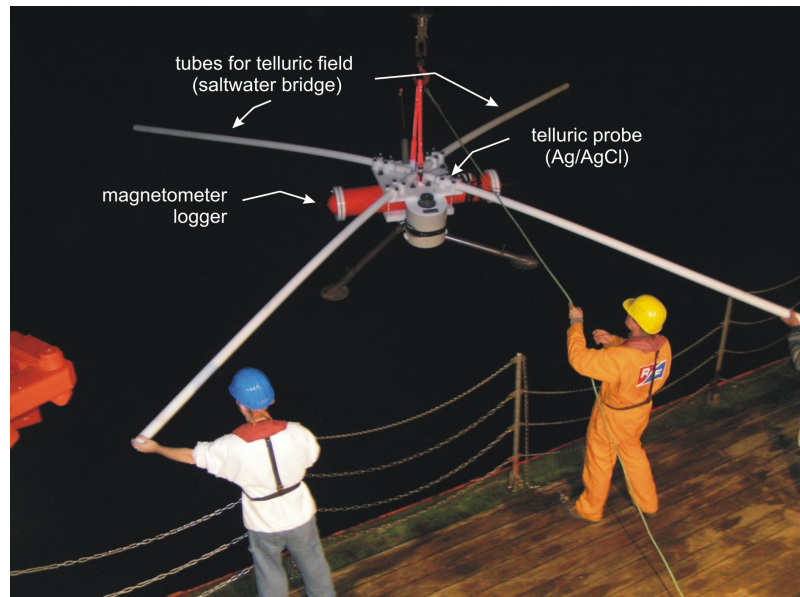


Fig. 2: Deployment of the HEFMAG seafloor magnetotelluric systems during the RV Sonne cruise SO181. The HEFMAG system consists of fluxgate magnetometer and short-span telluric device based on saltwater bridges.

The newly designed HEFMAG system (Woods Hole Oceanographic Institution) measures the natural temporal geomagnetic and geoelectric (telluric) field variations in a period range from $T = 80s - DC$.

While fluxgate magnetometers sense the three components of the magnetic field, a short-span device incorporating a saltwater bridge and unpolarizable Ag/AgCl electrodes is used for probing the electric field (Fig. 2).

Mainly due to the tubes for the telluric field, the ocean bottom instruments descends and rises at relatively low speed (approx. 20m/min).

For subsequent modelling the ocean has to be incorporated as an a priori structure in the finite difference mesh and therefore the knowledge of water conductivity is essential. A CTD log to depths until 4000 m was performed close to the measuring area (Fig. 3). The conductivity of seawater depends on pressure, salinity and temperature and has been investigated by Horner and Frysinger (1963). While the effects of pressure and salinity variations in the ocean are only 2.5% and 5%, respectively and are therefore negligible, the resistivity of oceanic water can change from 0.16 Ωm at 30°C and 0.33 Ωm at 1.0°C. Thus the temperature is the dominant factor controlling the electrical conductivity (Filloux, 1987).

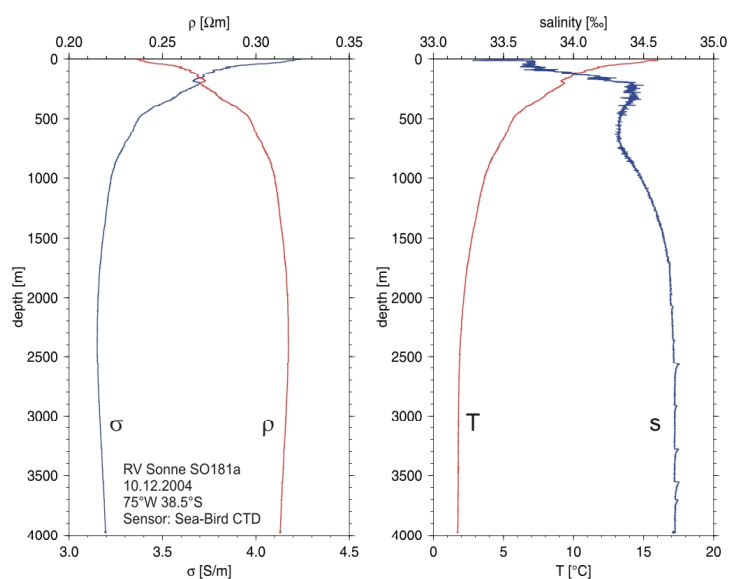


Fig. 3: Logs of electrical conductivity, resistivity (left), temperature and salinity (right).

Time series

The quality of the acquired electric and magnetic signals depends on the measure area. While the stations in the Coastal Cordillera and Volcanic Arc yield data of high quality, the electric field recorded by stations in Longitudinal Valley is strongly contaminated by cultural electrical noise, probably due to a higher density of population.

The offshore time series are attenuated at short periods. On the other hand, electrical induction by water motions (such as tides and ocean currents) could affect the time series at long periods. Fig. 4 shows examples of electromagnetic signals recorded on ocean bottom at 4163 m depth between December 2004 and February 2005. The data were rotated to geomagnetic North. In this plot daily variations and magnetic storms are evident as well as slow drift in the record of the electric field.

In Fig. 5 (left) a short section of the record is picked showing a magnetic storm in Jan 8, 2005. On the right side the time series of a running simultaneously land station is shown. Besides the high correlation between the on- and offshore fields one can clearly observe the filtering effect due to conductance of the ocean. High frequent signals (peaks) vanish nearly completely, because of the strong attenuation at short periods, giving the offshore-curves a smooth shape. Due to this impeding effect in propagation of electromagnetic fields, special care has to be taken with respect to signal-to-noise ratio and very long recording times are essential for obtaining high-quality transfer functions.

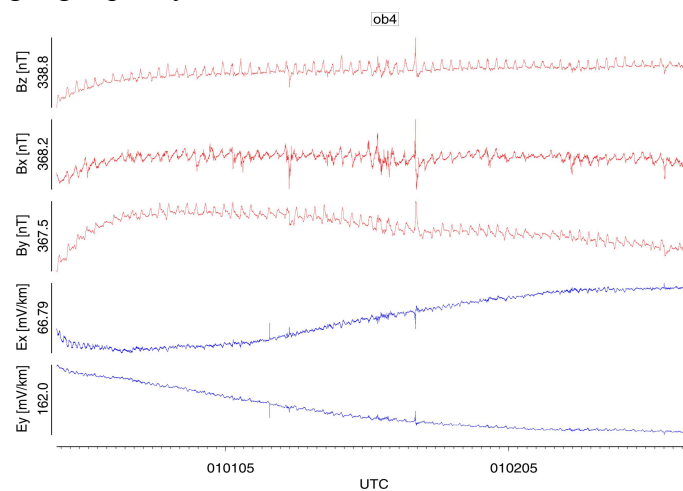


Fig. 4: Record of magnetic and electric field on ocean bottom in depth of 4163 m between December 2004 and February 2005.

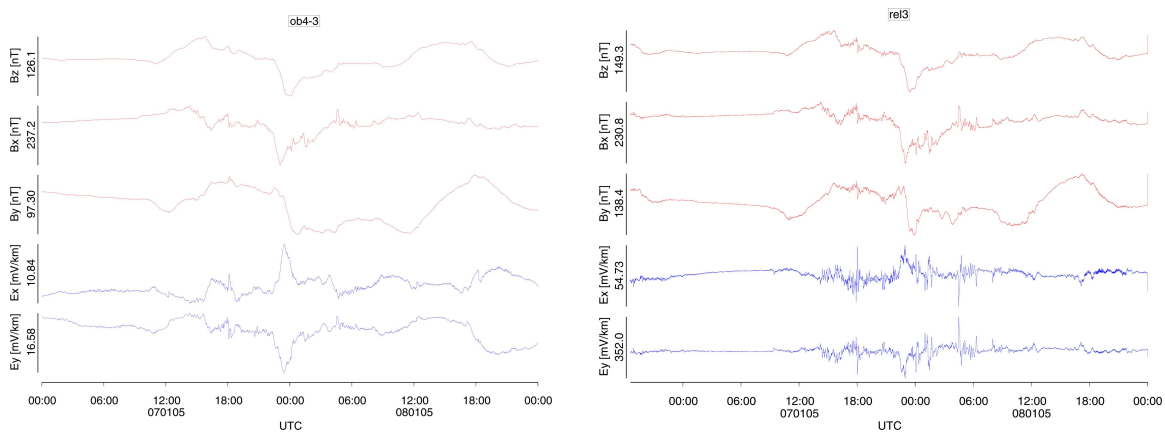


Fig. 5: Short section of time series at offshore site ob4 (see Fig. 4 above) and onshore site rel.

Transfer functions

Analysis of the time series have been carried out with the robust EMTF code of Egbert and Booker (1986). The code is adapted for single station and remote reference analysis and had to be modified for the ocean-bottom station data.

While analysis of offshore data is still ongoing, the remote reference processing of the onshore data yield predominantly smooth transfer functions in the period range of 10s – 10000s. However, in the Longitudinal Valley the contamination of the recorded electric field is responsible for perturbed transfer functions.

Examples of an onshore (left) and offshore (right) transfer functions are shown in Fig. 6. All land transfer functions are formed by multidimensional underground. Low values of the apparent resistivity are revealed in the eastern end of the profile. This indicates a body of enhanced conductivity in the lower crust east of the active volcanic arc and agrees with the behavior of the induction vectors.

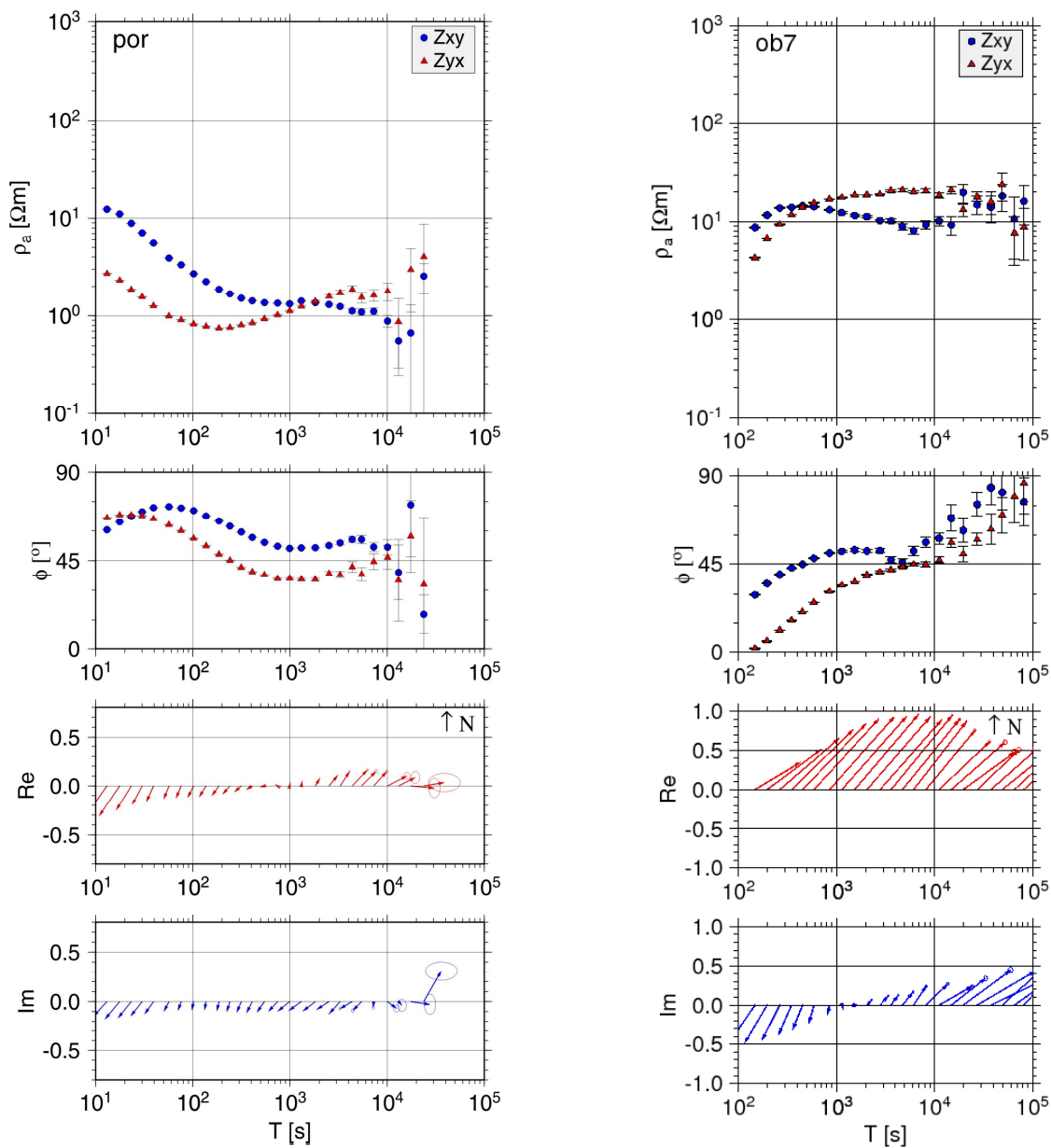


Fig. 6: Transfer functions of on- (left) and offshore profile (right).

Induction vectors

Fig. 7a and 7b show induction vectors from the previous and recent studies in the South Central Chilean continental margin for periods of 102 s and 3277 s.

Except for the westernmost stations, where the inductive influence of the Pacific ocean becomes visible in long vectors, all other vectors at 102 s are short and indicate no major conductivity contrast. In the eastern part of the area they point towards west and turn to east indicating increasing conductivities below the volcanic arc. Especially around Villarrica volcano, induction vectors point away from the center of the volcano and appear to be influenced by enhanced conductivities in the upper crust. However, the observed trend in the induction vectors can be caused by a conductivity structure like a magma chamber as well as perhaps by topography of the volcano edifice itself. But note that this effect isn't observed at Llama volcano. This suggests that a conductive structure beneath Villarrica volcano rather than topography produces the induction vectors.

A spectacular effect can be seen in the magnetic onshore transfer functions at long periods. Induction vectors were expected to point W-E near the Pacific coast and away from the volcanic arc. Instead, they point consistently to NE over hundreds of kilometres in the whole study area and their orientation seems to agree with the pattern of faults in South Central Andes. This could be explained by an anisotropy in the crust (see Brasse et al., this volume). This effect is not visible on the Nazca plate. The induction vectors are small and don't show any special orientation. This suggests that the geological and tectonical pattern of the Nazca plate differs from that of the South American one.

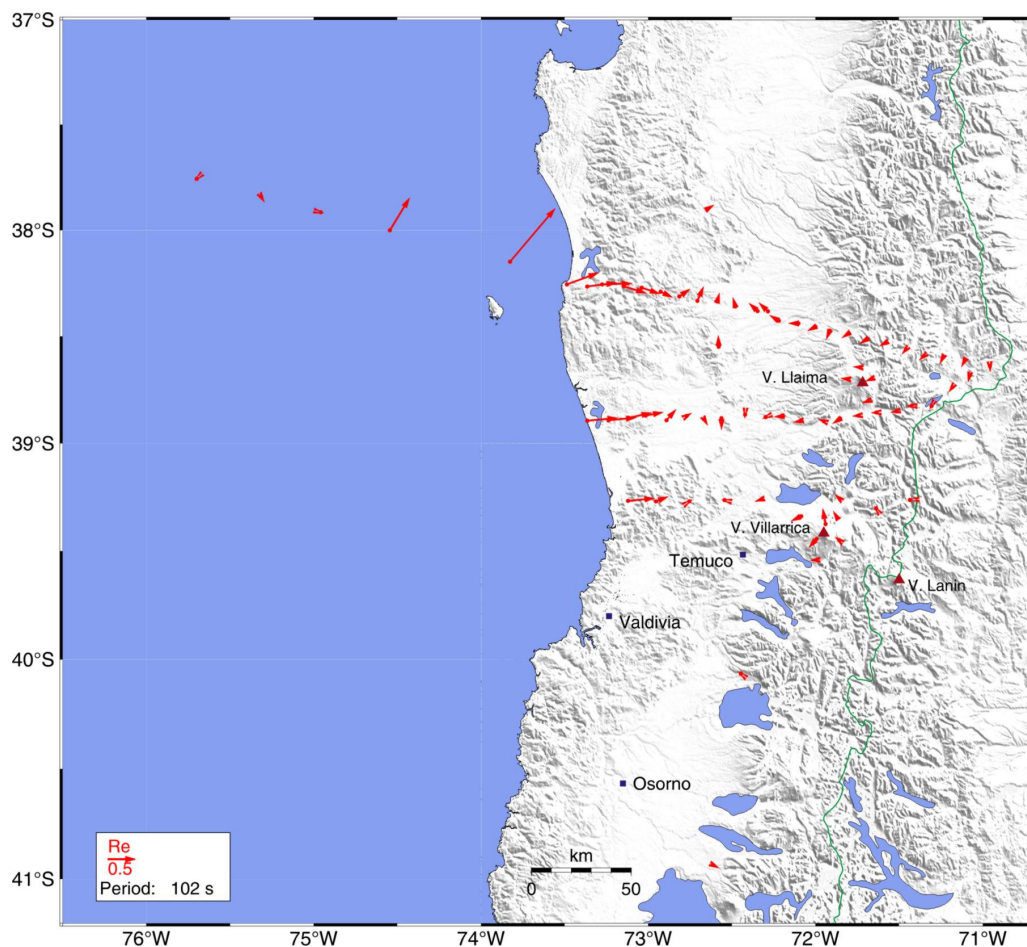


Fig. 7a: Induction vectors for 102 s from the previous and recent magnetotelluric profiles at South Chilean continental margin.

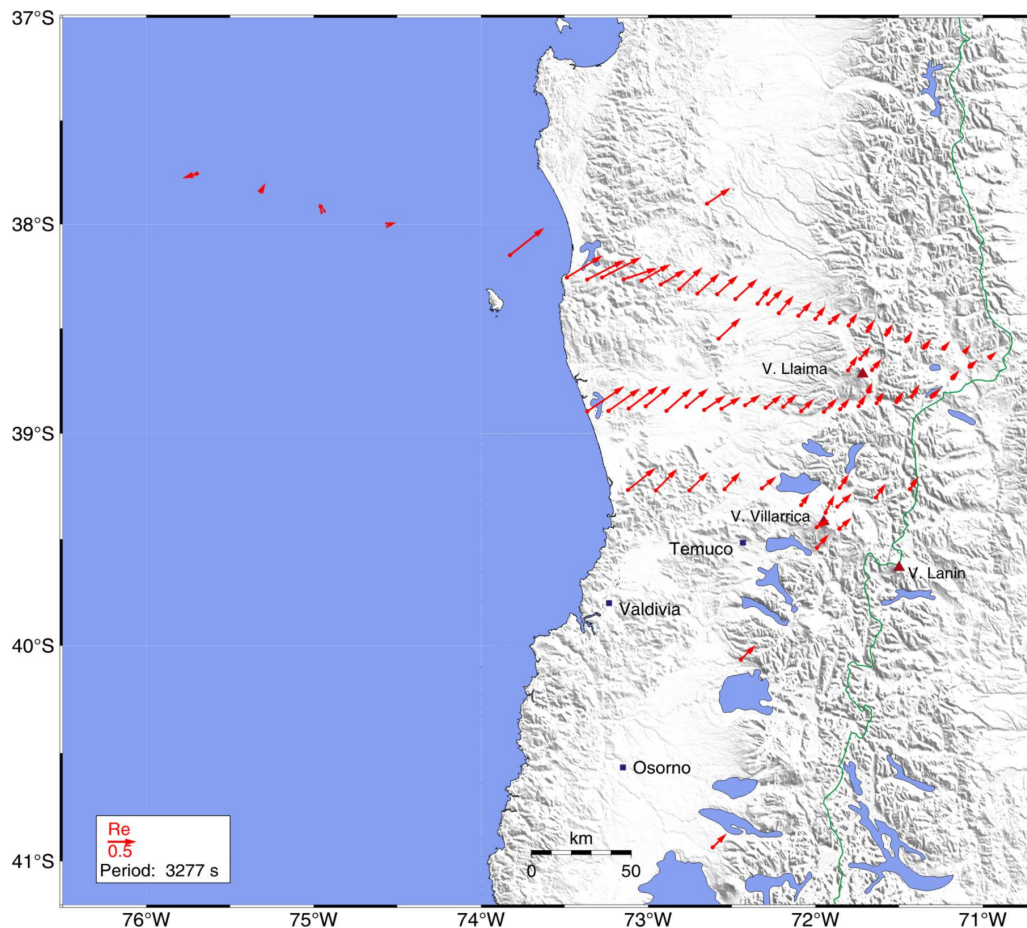


Fig. 7b: Induction vectors for 3277 s .

Discussion and Outlook

The first data analysis confirms the results of previous investigations. The NE orientation of the geomagnetic induction vectors on the continental plate is also visible in the recent profile. This feature could be reasonably reproduced by forward modelling incorporating anisotropy in the crust. The behavior of the induction vectors changes on the Nazca plate and reflects the geological and tectonical differences between these plates.

Unfortunately, the postulated model doesn't fit satisfactorily the measured magnetotelluric transfer function and requires further modifications.

Results of the previous studies reveal an enhanced conductivity in the lower crust below and east of the active volcanic arc and the question arises, whether the existence of such conductors can be confirmed beneath the current profile. In particular the inclusion of the offshore data in the modelling will show if the aim of the experiment can be achieved, which is the resolving of the good conductor on the interface between the plates.

However, simple isotropic 2D Inversion is not sufficient to produce a confident and stable model and has to be combined with the anisotropic model. The crustal anisotropy is a challenge for inversion, because it masks a deep view into the earth and thus complicates the understanding of the conductivity structure of the entire subduction zone.

References

- Brasse, H. and Soyer, W. (2001): A magnetotelluric study in the Southern Chilean Andes, *Geophys. Res. Lett.*, 28 (19), 3757-3760.
- Brasse, H., Li, Y., Kapinos, G., Eydam, D. and Mütschard, L. (2006): Uniform Deflection of Induction Vectors at the South Chilean Continental Margin: A Hint at Electrical Anisotropy in the Crust, this volume.
- Egbert, G. D. (1997): Robust multiple station magnetotelluric data processing, *Geophys. J. Int.*, 130, 475-496.
- Filloux, J. H. (1987): Instrumentation and Experimental Methods for Oceanic Studies, - In: J. A. Jacobs (ed.): *Geomagnetism*, vol. 1: 143-248; *Academic Press*, San Diego.
- Horner, R. A. and Frysinger, G. R. (1963): *J. Geophys. Res.* 7, 1967-1973.
- Melnick, D., Rosenau, M., Folguera, A., and Echtler, H.P. (2006): Neogene tectonic evolution of the Neuquén Andes western flank (37-39°S), in: Kay, S.M. & Ramos, V.A. (eds.): *Evolution of an Andean margin: A tectonic and magmatic view from the Andes to the Neuquén Basin (35-39°S)*, *Geol. Soc. Am. Spec. Paper*, 407, doi:10.1130/2006.2407(04).
- Soyer, W. (2002): Analysis of geomagnetic variations in the Central and Southern Andes, *Dissertation, Fachbereich Geowissenschaften, Freie Universität Berlin*, Berlin 2002.

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